

# Radiative Transitions of Strange Heavy Axial Mesons in Light-Cone QCD Sum Rules

S. Bilmis et al.

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## Abstract

We study the radiative transitions  $D_{s1}(2460) \rightarrow D_s\gamma$ ,  $D_{s1}(2536) \rightarrow D_s\gamma$ , and  $B_{s1}(5830) \rightarrow B_s\gamma$  with light-cone QCD sum rules. The calculation includes hard photon emission from both quark lines, two-particle photon distribution amplitudes, and three-particle photon distribution-amplitude corrections from the background-gluon term in the quark propagator. The strange-quark mass is retained explicitly. We compare a legacy magnetic-susceptibility photon input with a recent lattice normalization of the leading-twist photon distribution amplitude. For the charm-strange channels we include a physical-current residue normalization calibrated by local QCD sum rules, while for the bottom-strange channel we use Pullin–Zwicky basis-current inputs followed by a mixed-current diagonalization closure. The preferred lattice-normalized results give a moderate width for  $D_{s1}(2460) \rightarrow D_s\gamma$ , a strongly cancellation suppressed  $D_{s1}(2536) \rightarrow D_s\gamma$  width, and a sub-keV median prediction for  $B_{s1}(5830) \rightarrow B_s\gamma$ .

## 1 Introduction

Radiative decays of positive-parity strange heavy mesons provide a sensitive probe of their spin structure, electromagnetic couplings, and proximity to open-flavor thresholds. The charm-strange states  $D_{s1}(2460)$  and  $D_{s1}(2536)$  have long been discussed in this context, and  $D_{s1}(2460) \rightarrow D_s\gamma$  has an experimentally measured branching fraction [1]. The high state  $D_{s1}(2536)$  has a measured total width [2], but its radiative decay to  $D_s\gamma$  is expected to be small. The bottom-strange state  $B_{s1}(5830)$  is observed spectroscopically [3], while its radiative decay remains a target for future searches.

The purpose of this work is to update the light-cone sum-rule analysis of these radiative channels with a transparent separation of hard photon emission, two-particle photon distribution amplitudes (DAs), and three-particle photon DAs. We keep explicit  $m_s$  effects and track the normalization associated with the two physical axial currents. We also compare the older magnetic-susceptibility input for the photon DA [4, 5] with the recent lattice normalization of the leading-twist photon DA [6].

## 2 Conventions and Hadronic Representation

We use the mostly-minus metric,  $\epsilon^{0123} = +1$ , and the momentum assignment

$$P = p + q, \tag{1}$$

where  $P$  is the initial axial-meson momentum,  $p$  is the final pseudoscalar momentum, and  $q$  is the real-photon momentum. The pseudoscalar interpolating current is

$$J_5(x) = \bar{s}(x) i\gamma_5 Q(x), \tag{2}$$

with  $Q = c$  or  $b$ . For the axial channel we use the two basis currents

$$J_\mu^A(x) = \bar{s}(x)\gamma_\mu\gamma_5 Q(x), \quad (3)$$

$$J_\mu^B(x) = \frac{i}{m_Q + m_s} \bar{s}(x)\sigma_{\mu\alpha}P^\alpha\gamma_5 Q(x). \quad (4)$$

The tensor current therefore carries the initial-state momentum  $P^\alpha$ . The physical-current combinations are taken as

$$J_{1\mu} = \sin\theta J_\mu^A + \cos\theta J_\mu^B, \quad (5)$$

$$J_{2\mu} = \cos\theta J_\mu^A - \sin\theta J_\mu^B, \quad (6)$$

with central  $\theta = 35.3^\circ$  and a scan over  $25^\circ \leq \theta \leq 45^\circ$ .

The transition amplitude is parametrized by an E1 coupling  $G$ ,

$$\langle P(p)\gamma(q, \varepsilon) | A(P, \eta) \rangle = e G [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)], \quad (7)$$

so that

$$\Gamma(A \rightarrow P\gamma) = \frac{\alpha_{\text{em}}}{3} G^2 |\vec{q}|^3, \quad |\vec{q}| = \frac{m_A^2 - m_P^2}{2m_A}. \quad (8)$$

### 3 Light-Cone Sum Rule

The correlation function is

$$\Pi_\mu(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \Gamma_\mu Q(x) \bar{Q}(0) i \gamma_5 s(0) \} | 0 \rangle, \quad (9)$$

where  $\Gamma_\mu$  denotes either the axial or tensor basis current. We project onto the transverse structure

$$T_{\mu\nu} = (p \cdot q)g_{\mu\nu} - q_\mu p_\nu. \quad (10)$$

The OPE includes:

- hard photon emission from both the heavy and strange quark lines;
- two-particle photon DAs, including the leading-twist magnetic susceptibility or lattice-normalized transverse photon input;
- three-particle photon DAs generated by the background-gluon insertion in the quark propagator.

Ordinary vacuum-condensate terms are not inserted as separate corrections in the base propagators of the three-point photon LCSR. Condensate dependence enters through photon matrix elements and photon-DA normalizations such as  $\chi_s \langle \bar{s}s \rangle$ . This is distinct from a local two-point decay-constant sum rule, where ordinary SVZ condensates do appear.

After Borel transformation and continuum subtraction, the basis-current couplings are denoted by  $G_A$  and  $G_B$ . For the charm-strange physical currents we use

$$G_{2460} = \frac{\sin\theta f_A G_A + \cos\theta f_B G_B}{f_1}, \quad (11)$$

$$G_{2536} = \frac{\cos\theta f_A G_A - \sin\theta f_B G_B}{f_2}. \quad (12)$$

The residue model is calibrated with the working Stage-3 anchor  $f_1 = 0.345 \pm 0.017$  GeV; the same two-point overlap model predicts  $f_2 \simeq 0.38$  GeV. As a verification, we compare the mixed-current model with the pure axial-current local-QCDSR benchmark  $f_A(D_{s1}) = 0.245 \pm$

0.017 GeV from Wang’s Table 1 [7]. Since Wang uses the pure axial current  $J_A$ , the direct comparison is only to our limiting-current cases:  $J_1 = J_A$  at  $\theta = 90^\circ$  or  $J_2 = J_A$  at  $\theta = 0^\circ$ . Our working pure-axial input  $f_A = 0.225 \pm 0.025$  GeV is compatible with Wang’s value within uncertainties. At the physical angle, the Wang-like target requires a strongly negative  $AA/BB$  overlap and is therefore treated as a diagnostic benchmark, not as a direct determination of the physical mixed residues. The comparison is summarized in Table 2. For the bottom-strange channel we use the Pullin–Zwicky basis-current inputs,  $f_{B_{s1},A}^{\text{PZ}} = 0.305 \pm 0.027$  GeV and  $f_{B_{s1}}^{T,\text{PZ}} = 0.285 \pm 0.027$  GeV [8]. We then obtain the physical-current decay constants from a mixed-current diagonalization closure. For the observed high-combination state in the preferred lattice-photon row this gives  $f_1^{(B_s)} = 0.443[0.384, 0.492]$  GeV and  $f_2^{(B_s)} = 0.203[0.121, 0.291]$  GeV.

## 4 Numerical Inputs

Table 1 summarizes the main numerical inputs. Masses, quark masses, and electromagnetic normalization are taken from PDG summaries [9]; decay constants and lattice context are taken from FLAG and related heavy-light sum-rule literature [10, 11, 8, 7]. Photon DAs follow Ball, Braun and Kivel [4], the legacy magnetic susceptibility comparison follows Rohrwild [5], and the modern photon normalization scenario uses Bacchio et al. [6].

| Quantity                        | Symbol                                               | Value                                        | Source/use                                   |
|---------------------------------|------------------------------------------------------|----------------------------------------------|----------------------------------------------|
| quark masses                    | $m_c, m_b, m_s$                                      | 1.27(2), 4.18(3), 0.093(11) GeV              | PDG inputs, with explicit $m_s$ terms        |
| charm hadron masses             | $M_{2460}, M_{2536}, m_{D_s}$                        | 2.4595, 2.53511, 1.96835 GeV                 | PDG                                          |
| bottom hadron masses            | $M_{5830}, m_{B_s}$                                  | 5.82870, 5.36692 GeV                         | PDG/CMS                                      |
| lower $B_{s1}$ diagnostic mass  | $M_{B_{s1}}^{\text{low}}$                            | 5.750(26) GeV                                | lattice-inspired diagnostic [12]             |
| $D_s$ and $B_s$ decay constants | $f_{D_s}, f_{B_s}$                                   | 0.2499, 0.2303 GeV                           | FLAG/lattice context                         |
| charm physical residues         | $f_1, f_2$                                           | 0.345(17), $\simeq 0.38$ GeV                 | working Stage-3 anchor and predicted partner |
| pure axial benchmark            | $f_A(D_{s1})$                                        | 0.245(17) GeV                                | local-QCDSR comparison only                  |
| bottom basis constants          | $f_{B_{s1},A}^{\text{PZ}}, f_{B_{s1}}^{T,\text{PZ}}$ | 0.305(27), 0.285(27) GeV                     | Pullin–Zwicky                                |
| bottom physical constants       | $f_1^{(B_s)}, f_2^{(B_s)}$                           | 0.443[0.384, 0.492], 0.203[0.121, 0.291] GeV | this work                                    |
| legacy photon input             | $\chi_s(1 \text{ GeV})$                              | +3.15(30) GeV <sup>-2</sup>                  | comparison scenario                          |
| lattice photon input            | $f_{\gamma,s}^\perp(1 \text{ GeV})$                  | −55.1(1.9) MeV                               | preferred modern scenario                    |

Table 1: Main numerical inputs. The complete bookkeeping tables are `inputs_table.csv`, `outputs/paper_input_summary_table.csv`, and `outputs/input_citation_map.csv`.

| Case                             | Current/target               | $\theta$     | $\rho_{AB}$ | $f_A$     | $f_1$ | $f_2$ |
|----------------------------------|------------------------------|--------------|-------------|-----------|-------|-------|
| Wang benchmark                   | pure $J_A$ two-point current | —            | —           | 0.245(17) | —     | —     |
| our pure- $J_1$ axial limit      | $J_1 = J_A$                  | $90^\circ$   | 0           | 0.225     | 0.225 | 0.462 |
| our pure- $J_2$ axial limit      | $J_2 = J_A$                  | $0^\circ$    | 0           | 0.225     | 0.462 | 0.225 |
| physical angle, Wang-like target | target $f_1 = 0.245$         | $35.3^\circ$ | −1.00       | 0.225     | 0.247 | 0.451 |
| physical angle, Wang $f_A$       | target $f_1 = f_A = 0.245$   | $35.3^\circ$ | −0.957      | 0.245     | 0.245 | 0.462 |
| current Stage-3 anchor           | target $f_1 = 0.345$         | $35.3^\circ$ | −0.408      | 0.225     | 0.345 | 0.381 |

Table 2: Pure axial-current benchmark and limiting cases of the mixed-current residue model. Wang’s pure axial result is  $f_A(D_{s1}) = 0.245 \pm 0.017$  GeV; it tests the  $J_A$  scale through the pure-current limits, while the physical mixed-current residues require an assumption about  $\rho_{AB}$ . All listed residues are in GeV.

## 5 Results

The final form-factor and width predictions are shown in Table 3. The table quotes medians and 16–84 percentile intervals. Percentile intervals are preferred because several channels, especially  $D_{s1}(2536)$  and  $B_{s1}(5830)$ , are cancellation sensitive and non-Gaussian near zero.

| Sector | Decay                                 | Photon input                 | Mixing                      | Normalization                       | $G$ (GeV $^{-1}$ )          | $\Gamma$ (keV)            |
|--------|---------------------------------------|------------------------------|-----------------------------|-------------------------------------|-----------------------------|---------------------------|
| $D_s$  | $D_{s1}(2460) \rightarrow D_s \gamma$ | legacy $\chi_s$              | $35.3^\circ$                | $f_1 = 0.344[0.329, 0.363]$         | $-0.424[-0.484, -0.360]$    | $37.7^{+11.5}_{-10.5}$    |
| $D_s$  | $D_{s1}(2536) \rightarrow D_s \gamma$ | legacy $\chi_s$              | $35.3^\circ$                | $f_2 = 0.380[0.339, 0.423]$         | $+0.0118[+0.0014, +0.0233]$ | $0.045^{+0.124}_{-0.040}$ |
| $D_s$  | $D_{s1}(2460) \rightarrow D_s \gamma$ | lattice $f_{\gamma,s}^\perp$ | $35.3^\circ$                | $f_1 = 0.345[0.330, 0.364]$         | $-0.277[-0.312, -0.243]$    | $16.1^{+4.4}_{-3.7}$      |
| $D_s$  | $D_{s1}(2536) \rightarrow D_s \gamma$ | lattice $f_{\gamma,s}^\perp$ | $35.3^\circ$                | $f_2 = 0.379[0.338, 0.423]$         | $+0.0403[+0.0194, +0.0596]$ | $0.504^{+0.600}_{-0.377}$ |
| $B_s$  | $B_{s1}(5830) \rightarrow B_s \gamma$ | legacy $\chi_s$              | $35.3^\circ$                | $f_2^{(B_s)} = 0.218[0.138, 0.292]$ | $-0.0793[-0.130, -0.0537]$  | $1.33^{+2.27}_{-0.72}$    |
| $B_s$  | $B_{s1}(5830) \rightarrow B_s \gamma$ | lattice $f_{\gamma,s}^\perp$ | $35.3^\circ$                | $f_2^{(B_s)} = 0.203[0.121, 0.291]$ | $-0.0401[-0.0695, -0.0274]$ | $0.342^{+0.684}_{-0.182}$ |
| $B_s$  | $B_{s1}(5830) \rightarrow B_s \gamma$ | legacy $\chi_s$              | $25^\circ\text{--}45^\circ$ | $f_2^{(B_s)} = 0.246[0.174, 0.312]$ | $-0.0380[-0.103, +0.0101]$  | $0.349^{+1.908}_{-0.323}$ |
| $B_s$  | $B_{s1}(5830) \rightarrow B_s \gamma$ | lattice $f_{\gamma,s}^\perp$ | $25^\circ\text{--}45^\circ$ | $f_2^{(B_s)} = 0.249[0.170, 0.307]$ | $+0.0137[-0.0612, +0.0657]$ | $0.493^{+1.239}_{-0.441}$ |

Table 3: Final form-factor and width summary. The lattice photon input is our preferred modern normalization; the legacy susceptibility input is kept for comparison with older photon-DA analyses.

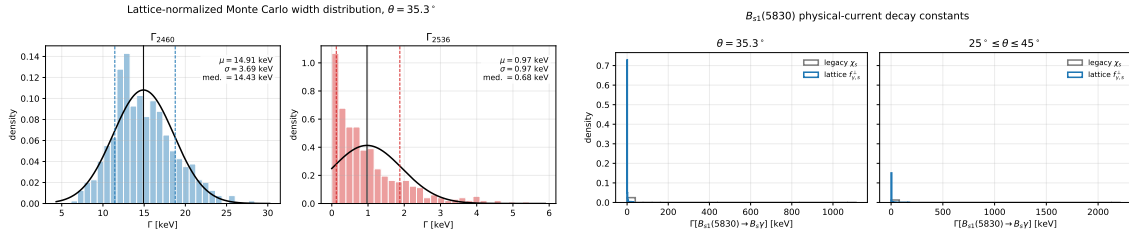


Figure 1: Monte Carlo width distributions for the charm-strange lattice photon-normalization scenario (left) and the bottom-strange physical-current normalization (right).

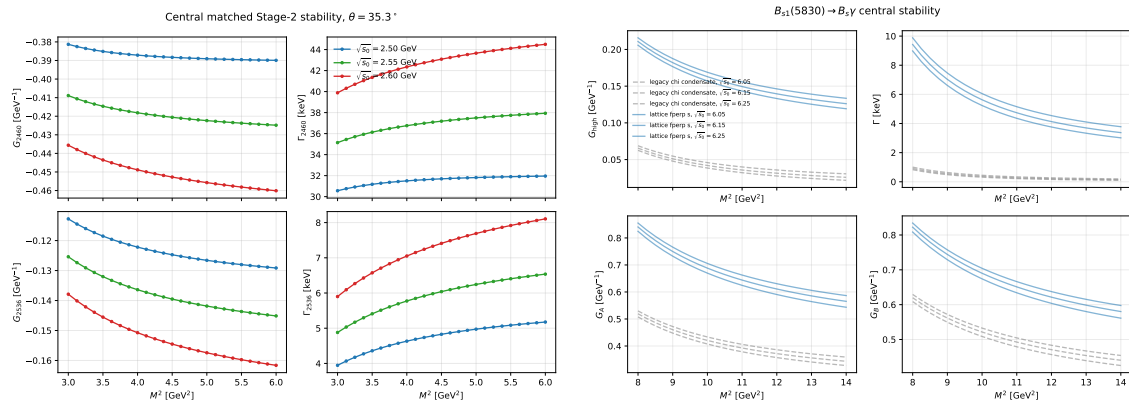


Figure 2: Representative stability plots in the selected Borel and continuum windows for the charm-strange and bottom-strange analyses.

## 6 Comparison with Experiment

For  $D_{s1}(2460)$ , BaBar measures a radiative branching fraction  $\mathcal{B}(D_{s1}(2460) \rightarrow D_s \gamma) = (16 \pm 4 \pm 3)\%$  [1]. The preferred lattice-normalized partial width,  $16.1^{+4.4}_{-3.7}$  keV, corresponds to a total width of order 0.10 MeV if this branching fraction is used literally. This is compatible with the observed narrowness of the state.

For  $D_{s1}(2536)$ , using the measured total width  $\Gamma_{\text{tot}} = 0.92 \pm 0.05$  MeV [2], the preferred lattice-normalized radiative width gives an estimated branching fraction of order  $5 \times 10^{-4}$ . The channel is therefore strongly cancellation suppressed, which is consistent with the lack of a prominent experimental signal.

For  $B_{s1}(5830)$ , there is no direct radiative measurement. Our physical-current-normalized prediction is a future-search target. In the preferred lattice-normalized scenario the median width remains below 1 keV, although the mixing-angle scan leaves a long upper tail.

## 7 Conclusions

We have constructed a staged light-cone QCD sum-rule analysis of  $D_{s1} \rightarrow D_s \gamma$  and  $B_{s1} \rightarrow B_s \gamma$  radiative transitions. The calculation includes hard photon emission from both quark lines, two-particle photon DAs, three-particle photon DAs, and explicit strange-quark-mass terms. The modern lattice normalization of the leading-twist photon DA reduces the dominant photon-input ambiguity relative to the older susceptibility-only scenario. The resulting pattern is physically clear:  $D_{s1}(2460)$  can have a visible radiative mode while  $D_{s1}(2536)$  and the observed high-combination  $B_{s1}(5830)$  are cancellation suppressed.

## Acknowledgments

To be added.

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